KAPASA MAKASA UNIVERSITY

AGRICULTURAL ENTOMOLOGY

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UNIT TWO

TOPIC: APPLYING SYSTEMATICS AND TAXONOMIC TECHNIQUES

Objectives: by the end of this unit, SSBAT;

- Explaininsect systematics,diversity and evolution
- Carry out identification and classification of insects
- State the importance of identification and classification of insects.

2.1. Explaining insect systematics, diversity and evolution

2.2.1. Insect Systematics

Insect systematics or systematic entomology is the study of the diversity of insects and their interrelationships. Systematics can be subdivided into two fields' which are **taxonomy** and **phylogenetics**. Insect taxonomy is the science that deals with recognizing, describing and naming species and groups of species, and the classification of insects into a ranked and named system (e.g., species, genus, family, order, etc.) that aims to reflect a natural evolutionary history. Phylogenetics is the study of the relatedness of organisms or groups of organisms based on shared ancestry.

2.1.2. Evolutionary history of insects

Insects are the most diverse and important group of animals on land. There are more insects on land than all the other animals put together. Insects represent approximately 85% of all known animals. There is close to 400lbs. of insect biomass as compared to 14lbs. of fresh and blood. The population of insects to other animals is said to be 4:1. Insects are very successful inhabitants of the earth and in anything living (*everywhere*), mainly due to the following characteristics they possess:

1. They have **tremendous reproductive potential**;
2. They have **high survival capacity**;
3. They are **small in size**;
4. They are **diverse;**
5. They have **short life cycles.**
6. They have **adaptation appendages**.

Insects can exist as predators, parasites, hosts, or herbivores or decomposers. They are invertebrates (*a group of animals without internal skeleton*) and belong to a larger group called arthropoda. The unique features of arthropods are that their body parts are divided into three distinct parts (**tagmata**) namely; Head (*has feelers or antennae*), Thorax (*chest with 2pairs of wings and 3pairs of legs*) and Abdomen (*has spiracles*). Other groups to which arthropods are classified are;

- Crustacea
- Arachnida
- Insecta
- Myriapoda

a. Crustacea – have two pairs of antennae and under this are shrimps, lobsters, crabs etc.
b. Arachnida – have no antennae, has head and thorax joined together and usually have four (4) pairs of legs e.g. spider, mite and scorpion.
c. Insecta – have one pair of antennae, two pairs of wings and usually three pairs of legs e.g. bees, grasshopper and beetles.
d. Myriapoda – myriapods have several pairs of legs and their abdomen and thorax not clearly distinguished e.g. centipeds and millipedes.

Arthropods usually have a rigid exo-skeleton and the legs are jointed, hence the term arthropod refers to jointed foot. The word ‘‘Arthro”in greek means a joint while ‘‘podos’’ means foot. In order for arthropods to grow, they have to shade the external skeleton all at once through the process referred to as moulting.

2.1.3. Describing insect abundance

Insect abundance is the estimated number of individual insect species per unit area. It is expressed by assigning a species to one of the following classes:

Abundance Classes

CLASSES	**STOCKS PER M^2 QUADRAT**
Rare	1 to 4
Occasional	5 to 14
Frequent	15 to 29
Abundance	30 to 99
Very abundant	100+

The abundance refers to density of population in those quadrats in which a given species occur. This can be recorded by insect counts in the quadrats.

2.1.3. Examining insect diversity

The simplest measure of density is to count the total number of species in an area. This also termed as species richness. Species richness, however, does not take into account species abundance (refer to table below).

COMMUNITY TYPE	# OF INDIVIDUALS OF SPECIES 1	# OF INDIVIDUALS OF SPECIES 2
A	99	1
B	50	50

The species richness (diversity) of community B, with two species of population of 50 each, will equal that of A in which the populations of species are 99 and 1.

Recorded Surveys

The canopy of the rainy forest trees are believed to hold the key to the immerse diversity of insects. The unknown range of plant host –specificities of tropical insects is the main weakness in insect diversity count. The preliminary result of such count suggest that over 42 million arthropods may be found in a hectare of an area of a rain forest (at the time of this study), and that 70% occur in the soil and leaf litter and 14% in the canopy (Stock, 1988). At the 1977 Royal Entomological Society's symposium on insect diversity, a pie chart was presented representing the relativecontributions of the described species of plants and animals (Southwood, 1978) and this showed the striking dominance of the insects. Despite other later presentations, the proportion of described animals and plants species still remain largely the same, thus insect still being dominant as shown in the pie-chart below:-

Getting the measure of diversity

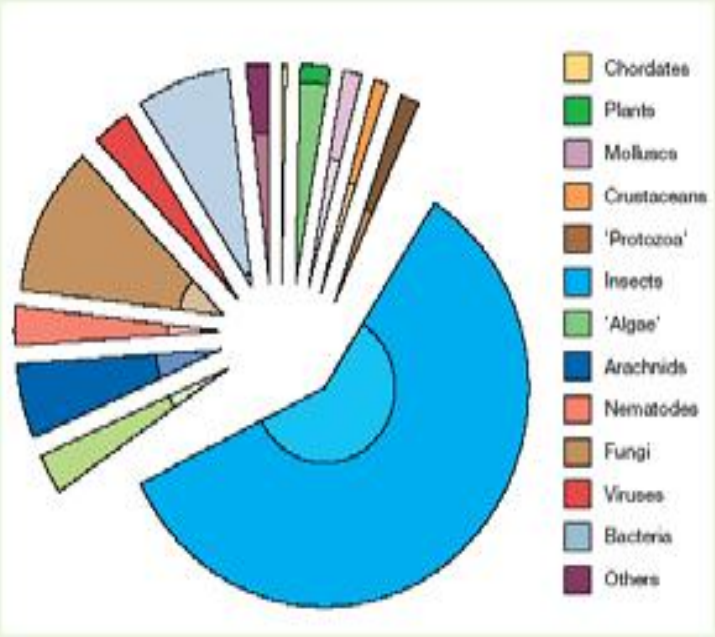


Figure 4 Species richness in major groups of organisms. The main 'pie' shows the species estimated to exist in each group; the hatched area within each slice shows the proportion that have been formally described. Data from ref. 7.

2.2. Carrying out insect identification and classification

2.2.1. Binomial nomenclature

The system of binomial nomenclature we use today for classification and identification was advanced by Swedish naturalist Carolus Linnaeus, who first used it consistently in 1758. Strict rules and conventions apply for name assignment and for zoology. These are stated in the International Code of Zoological Nomenclature. Although they may be used in any language, scientific names are Latinized and usually refer to some characteristics of animals or a group named. The binomial of a species is always printed in italics or, if hand written, is underlined to indicate italics. The names of the genera and higher categories begin with a capital letter, while the specific epithet of the species and the sub-specific names always begin with the lowercase letter, as in the subspecies *Diabrotica undercimputata mannerheim* in this case the third name indicates a sub species, a group of individuals of the same species that have differences in body form or colour and geographical distribution.

The frequently used name for the species, the common name, is not covered by the formal rules of nomenclature. The name 'Gypsy moth'is a common name, somewhat similar to nick name. Such names are often used in insect pest management because they are easily pronounced compared with the Latin names. However, because no formal rules govern these names, we may find different names or versions used in different localities for the same species and confusion may arise. Because there is only one scientific name, it is always the safest name to use to avoid problems of semantics.

The species category plays the central role in systems of classification. It is the only category that is real; in other words, it is the only natural group. The species is a complex unit that is

sometimes difficulty to define, but a widely used definition of applied tothe species concept is that it is a group of interbreeding individuals that are similar in body structure and that produce fertile offspring; moreover, these groups are reproductively isolated from other such groups. This is a biological definition of species.

All other categories for the hierarchy are based on the species category but are just abstracts. Even though they are derived from suspected pathways of insect evolution, these other categories are mare human contrivances used for convenience. Because this is so, we find many differences in opinion as to where to lump and split categories, therefore, many different schemes of insect classification results. Consequently, there are no correct and non-incorrect systems. Rather we have successful and unsuccessful systems based on their usefulness.

2.2.2. Identification

Identifying insects is largely a process of patient observation. The process essentially involves the observation going about collecting and reviewing information, perhaps comparing it to other field notes – and finally, delivering the findings. During the process, consider very much what the insect do in the ecosystem. **There are three basic research items to consider about insect identification.**

2.2.2.1. Habitat.

Pay close attention to the environment of your specific insect. If you are looking for a certain type of insects understand where you can find them in nature e.g. gardens are tremendous source of finding many active species of spiders, butterflies, bees and beetles. An open field is another source for crickets as well as spiders. Forests and marshy areas attract specialized species that will be in abundance due to the water situations e.g. mosquitoes and dragon flies.

2.2.2.2. Time of the day

For a good portion of the insect Kingdom, activities usually are at the peakaround midday typically when the temperature is at its highest (especially true honey bees and butterflies). Some insects will become more active at dawn or dusk (such as mosquitoes) but most are busiest at night (spider and moth).

2.2.2.3. Equipment (optional)

Magnifying glass, digital camera, camera phone etc.

Once you have the above mentioned, then there are some questions that you need to ask yourself when you are trying to identify the insects such as;

How many legs does my insect have?

If the answer is six legs, you are most likely looking at an insect. If the answer is eight legs, you are most likely looking at a mite or spider.

Does the insect have any wings?

The answer to this question will tell you if it is a walking insect or a flying insect. Some insects do have wings but these prove unsuitable for flying especially over long distances. As such, the insect may resort to hopping about.

Does the insect have any noticeable antennae or feeler appendages?

If so do they end in point or are they knotted at their ends?

Are there any noticeable moving jaws (mandibles) or mouth parts?

This will tell you much about the types of food the insect may eat. Spiders will usually have biting pincer –like mouth parts whereas a butterfly will have a straw like mouth appendages for sucking nectar from flowers.

Taking these questions/ items into consideration and coupling them with observation techniques and an accurate insect resource (field guide) – one can begin to discern the varied amounts of insects' species in our world. When one finally gains a respect for these creatures, they will open their eyes to a brand new world that exists beyond our own comfortable lives.

2.2.3 Insect classification

In biology, the naming of organisms is referred to as nomenclature and ordering them into a hierarchy of categories is known as classification. Taxonomy involves the theoretical basis for classification.The classification of organisms is based on a hierarchy of categories, with the most inclusive occurring at the top and the least inclusive at the bottom. The major categories used in animal classification are the Phylum, Class, Order, Family, Genus and Species. But for added distinction in large, diverse groups, many other categories fall between these major ones. For example, a sub-class category is commonly present below the class category and a super family category above the family category. An example of the major categories for the Gypsy moth shows the following classification;

Phylum – Arthropoda

Class – Insect

Order – Lepidoptera

Family – Lymantridae

Genus – Lymantria

Species – *Lymantria dispar*

Note that each category consists of only one word except the species category. The scientific naming of species is binomial; it is composed of two names, a genus name and specific name, also known as a specific epithet. The specific name cannot stand alone but it must be used with the genus name.

General classification

Cleary, no one person can know all the kinds of insects in our environment. Not even trained entomologists instantly recognize every insect specimen presented before them. However, a good working knowledge of the major insect groups is desired with some degree of reading and laboratory study. The usual level of insect identification for communication with specialists is the **order** category. There are thirty one (31) orders of which eight (8) orders contain more than 95% of speciesand about seven (7) of these are the most diverse and account for the greatest share of pest problems in the forestry cycles and these include the following:

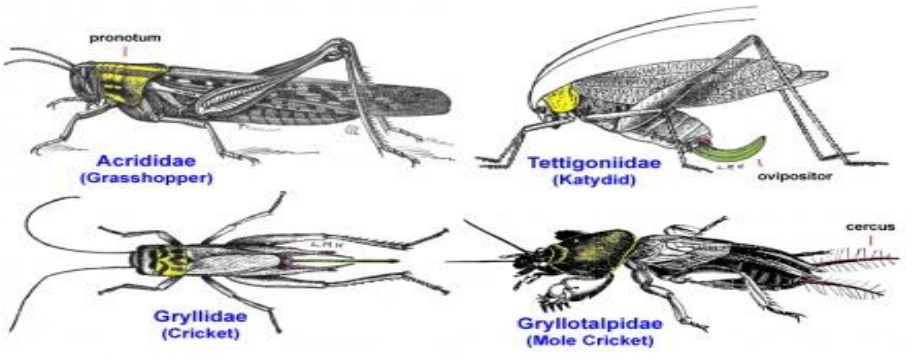
1. Orthoptera – grasshoppers, locusts and crickets
2. Homoptera – aphids, scale insects, hoppers, psyllids and whiteflies.
3. Hemipteran – bugs
4. Coleopteran – beetles
5. Diptera – flies and mosquitoes
6. Lepidoptera – moths, butterflies and skippers
7. Hymenoptera – bees, wasps and sawflies
8. Isoptera – termites

Orders closely related to Forestry

Orthopteran (grasshoppers, locusts and crickets)

This is the first of the major orders that contain important pest species.

- Although some are wingless, most have two (2) pairs of wings, with parchments like front wings, the tegmina, and membranous fanlike hind wings. These wings are folded over the back when not in use.
- Mouth parts are typical chewing type and have a pair of short growth (cerci) at the top end of the abdomen (grasshopper).
- They undergo gradual/ incomplete metamorphosis hence exhibit little change in body form during growth and development from larva to adult.
- Undergo one generation each year in temperate climate and the winter is usually passed in egg stage.
- Produce soundsthrough vibration of body parts, the behaviour called **stridulation**(grasshopper and crickets), usually involved in mate finding.

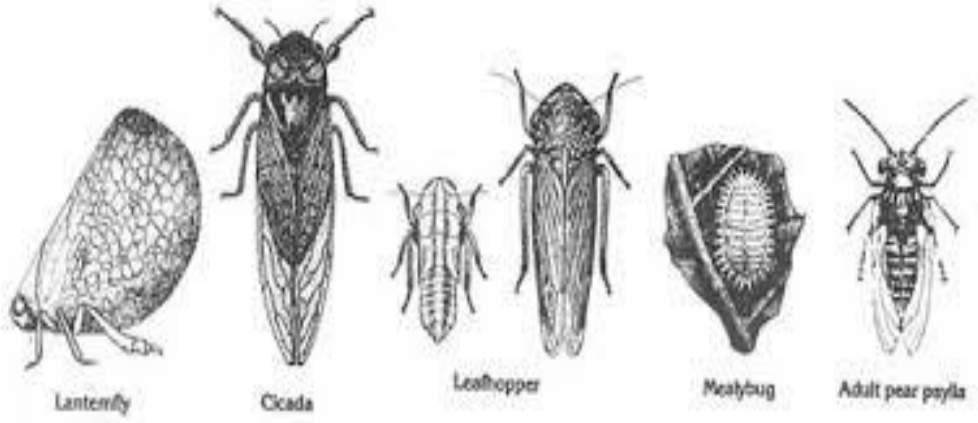


Hemiptera (bugs)

Colloquially, all insects are often referred to as **"bugs"** in America, however, in entomology; the word "bug" is strictly used for insects belonging to the order Hemiptera.Although insects of the family Coccinellidae are sometimes called ladybirds or ladybugs, these are neither birds nor true bugs, but actually beetles, and thus the names ladybird beetles or lady beetles are preferred by entomologists.For common names of true bugs, e.g., assassin bug, leaf-footed bug or giant water bug, the word 'bug' is written separately from its adjective or descriptor.

Most bugs are important pests but several of them are important natural enemies also that help to destroy insect pests. Some bugs are ecto-parasites of humans and other warm blooded animals. Bugs are some of the most common insects which occur in terrestrial and aquatic habitats.

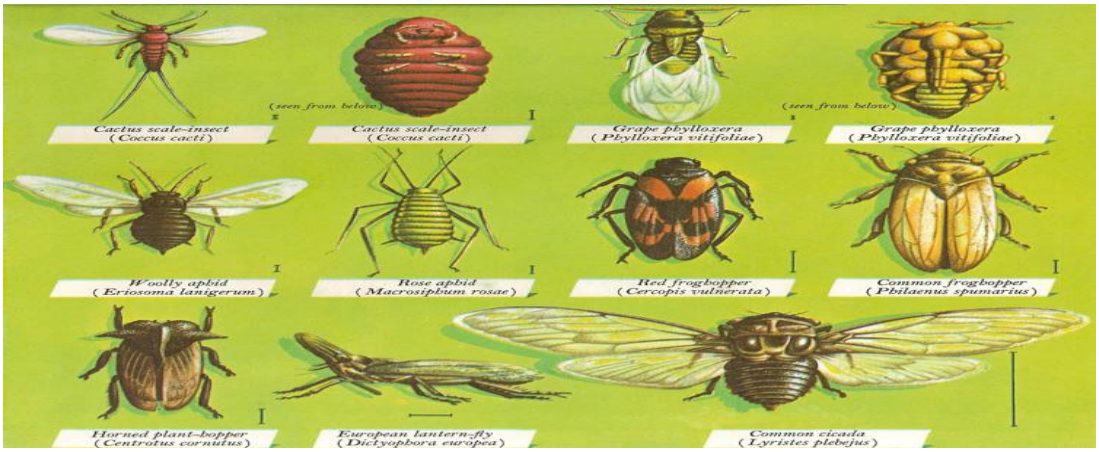
- Bugs have a flattened body, and their abdomen broadly joined to the thorax.
- Most have two (2) pairs of wings, the front wing called **hemelytra** are the most distinctive characteristic of the order.
- They generally have a long antennae and a typical piercing-sucking mouth with which they pierce tissues of plants or animals and extract sap or blood.
- Many have scent glands from which they secrete disagreeable odours, presumably for protection.
- Hemipteran are mostly found in plant tissues, on the surface of plants, or in crevices.
- They undergo a gradual growth and development metamorphosis, with the immature resembling adults but lacking fully developed wings and sometimes adult coloration.



Homoptera (aphids, scale insects, hoppers, psyllids and whiteflies)

The homoptera is a very large group of insects closely related to the Hemiptera. In fact, some authority groups the homoptera as a sub order of the hemiptera. They are so diverse that no single common name represents the whole order.

- Homopteran mouth parts are piercing sucking like those in the hemiptera, but unlike the hemiptera, wherein the beak arises from the front of the head, the homopteran beak joins to the back of the head.
- They have two pairs of wings but some are wingless at some stage, the front pair is membranous (or slightly thickened) and uniformly thick throughout, whereas back pair is thinner and membranous.
- They have well developed compound eyes e.g. aphids.
- The antennae vary from short and bristle like to long and threadlike.
- Homopterans are plant feeders and many are important agroforestry pests.



Coleoptera (beetles)

Beetles are by far the largest insect order - with more than one million described as pest species. This is the most important pest group in forestry. Beetles are highly diverse and exist as plant feeders and predators.

- As predators they serve as natural enemies and very useful in biological control of insects.
- They have chewing mouthparts and undergo complete metamorphosis.
- Most beetles possess two pairs of wings, the front pair is specialized and the most unifying characteristic of the order. These wings are thickened to form leathery or hard covering called elytra.
- When the insect is not flying, these elytra are folded to form a protection cover over the back. The hind wings are membranous and used for flight. They fold under the elytra when not in use.
- They have one pair of antennae.

- They are the most important pest group from forestry point of view.



Hymenoptera (bees, wasps, sawflies, ants, wood wasps and parasitic wasps)

This is the second largest insect order and considered by many people to be the most beneficial group in forestry. This view is based on activities of hymenoptera in plant pollination, in producing honey and bees wax, and in killing pest insects. However, certain sawflies, ants and other forms are pests and can be quite destructive.

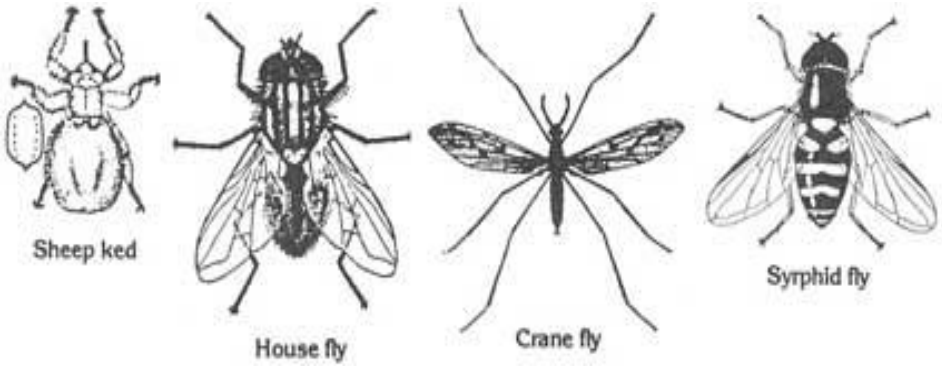
- These have mouth parts of chewing type but are modified for sucking or laying in highly evolved forms, such as bees.
- Most hymenopteran females have well developed ovipositor, and in some groups, it has been modified to form an effective stinging organ used for protection.
- Development in the hymenoptera occurs by complete metamorphosis, and immature (larvae) have well developed head capsules and chewing mouth parts.
- On attaining full growth, the larvae, sometimes, build silken or parchment cocoons.
- Most beneficial pests as plant pollinators (produce honey and bees wax) and, as natural enemies kills other insect pests



Diptera (flies and mosquitoes)

The fourth largest order of insects. Many insect common names end with the word **"fly"**, e.g., butterfly, dragonfly, mayfly, whitefly, but these are really not true flies; butterflies belong to the order Lepidoptera, dragonflies belong to the order Odonata, mayflies belong to the order Ephemeroptera and whiteflies belong to the order Hemiptera. In entomology, these are conventionally designated by writing the word fly in combination with a descriptor to form one word."True flies" are insects belonging to the order Diptera, and for this group, common names are usually written with the word fly separately, e.g., fruit fly, vinegar fly, deer fly, house fly. These are very diverse and successful insects that occurs in every conceivable terrestrial and fresh water habitats.

- As true flies, the dipterans are easily distinguished from the rest of the insects on the basis of their wings.
- They have only one pair of membranous wings arising from the meso-thorax and the hind pair has been reduced to form balancing organs called **halters.** Wings are absent in some Dipterans, but halters are usually present.
- Dipterans have diverse types of mouth parts e.g. mosquitoes have piercing-sucking mouth parts while house flies and related insects have sponging mouth parts.
- Development is by complete metamorphosis, and the immature take on a variety of shapes and sizes.
- Some Dipterans are important natural enemies of insects and others are considered as beneficial plant pollinators.
- Ordinarily not major pests of forests but fruit flies have been known to cause damage to some tree fruits.
- Most of them are important vectors of pathogens causing human diseases and domestic animals.



Lepidoptera (moths, butterflies and skippers)

Third largest insect order and second important pest group in forestry. They are less diverse than Beetles. Almost all are plant feeders as larvae while the adults feed mostly on nectar (some do not feed).There are more than 150,000 spp described so far. Many schemes of classification exist for the Lepidoptera. A more formal method divides the order into two major groups as (1) macro lepidoptera and (2) micro lepidoptrea according to average body size of included species. Another informal scheme divides the lepidoptera into moths and butterflies. Usually moths are distinguished from butterflies by having wings folded roof like or wrapped around the body when at rest while butterflies hold their wings vertically when at rest. Behaviorally, moths are active at night and butterflies are usually active during the day

- They undergo complete metamorphosis and in most cases the caterpillar stage is the most damaging and most preferred in eating.
- Caterpillars haveshort antennae, simple eyes and chewing mouth parts.
- The adults have three(3) pairs of jointed legs.
- Some species have well developed silk glands from modified salivary glands from which they build cocoons. These are shelters built by fully-grown moth caterpillars for transformation to adult stage. However, most butterfly caterpillars do not build cocoons.
- Lepidopteraare an order of great concern to pest management as almost all plant species have at least one caterpillar pest feeding on them.



The silverspot butterfly, *Dione juno*(Lepidoptera: Nymphalidae). Left. Larvae. Center. Pupae. Right.

Isoptera (termites)

Termites are a group of insects consisting of 2,500 species of which 300 are considered pests. Termites are one of the most damaging pests in the tropics and can cause considerable problems in agriculture, forestry and housing. The occurrence of termites can easily be noticed by the presence of termite moulds or termitaria. Termites are pests of considerable concern in forestry and cause serious damage to Eucalyptus spp. However, *E. tereticornis* and *E. coloeziana* show some resistant to their attack. Under this order, termites have been divided into three (3) main sub-groups, namely; Rhinotermitidae, Kalotermitidae and Termitidae.

- Rhinotermitidae - these are damp-wood termites, they attack wood with moisture content below 15% M.C.
- Kalotermitidae – these are non damp –wood termites, they attack wood with moisture content above 15 % M.C.
- Termitidae – are the majority of the termite species, about 80% of them fall in this group and attack heavily many plant species.
- Termites undergo a gradual growth and development metamorphosis, with the immature resembling adults but lacking fully developed wings.



The subterranean termite, *Reticulitermesflavipes*(Blattodea: Rhinotermitidae), a common structural pest in the southeastern USA.

Termites are also beneficial insects in the following ways:

- Nutrient recycling- breakdown and release of minerals from organic matter.
- Help in soil aeration-due to burrowing activities of termites.
- Food chain – an excellent source of proteins for many organisms such as ants, guinea fowl and other mammals including humans.
- Improved soil fertility - when termite mounds, which are rich in minerals, are crushed down and incorporated into the soil.